Sample Size in the Population Proportion Case.

Q: How do we figure out the necessary sample size w/ a required margin of error m?

$\longrightarrow$: pt. estimate $\pm$ margin of error

$$\hat{p} \pm \boxed{z^* \cdot \text{stderror} \quad \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}$$

w/ z^* the critical value of $N(0,1)$ for the confidence level C

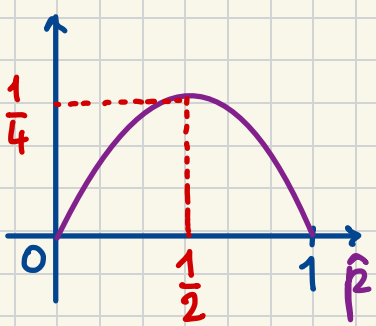
$$z^* \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq m \quad \Big/^2$$

$$(z^*)^2 \cdot \frac{\hat{p}(1-\hat{p})}{n} \leq m^2$$

$$(z^*)^2 \cdot \frac{\boxed{\hat{p}(1-\hat{p})}}{m^2} \leq n$$

We cannot know $\hat{p}$ before knowing n.

Consider $\hat{p}(1-\hat{p})$ as a function of $\hat{p}$.



The conservative choice for $\hat{p}$ is $\frac{1}{2}$.

$\Rightarrow$ The conservative choice for the sample size is

$$\boxed{n \geq \left(\frac{z^*}{2m}\right)^2}$$

Confidence Intervals for the Variance.

Consider a normal random sample $(Y_1, \dots, Y_n)$ w/ both parameters μ, σ unknown

A $\boxed{\text{good}}$ (unbiased) estimator for σ^2 is

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (Y_i - \bar{Y})^2$$

Theorem. In the above setup, set

$$Q^2 = \frac{1}{\sigma^2} \sum_{i=1}^{n} (Y_i - \bar{Y})^2$$

$$\tilde{Q}^2 = \frac{1}{\sigma^2} \sum_{i=1}^{n} (Y_i - \mu)^2 \sim \sum_{i=1}^{n} \left(\frac{Y_i - \mu}{\sigma} \right)^2 \sim \chi^2(df = n)$$

$\sim N(0,1)$

Then,

- $Q^2 \sim \chi^2(df = n-1)$
- Q^2 is a pivotal quantity for σ^2.

Problem 16.3. *What is the unbiased estimator for σ^2.*

$$\boxed{S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2}$$

Problem 16.4. *Assume a random sample $Y_1, Y_2, \ldots, Y_n$ from a normal distribution with mean μ and standard deviation σ - both unknown. What's the distribution of*

$$Q^2 = \frac{\sum_{i=1}^{n}(Y_i - \bar{Y}_n)^2}{\sigma^2} = \boxed{\frac{(n-1)S^2}{\sigma^2}}?$$

$$\boxed{\chi^2(df = n-1)}$$

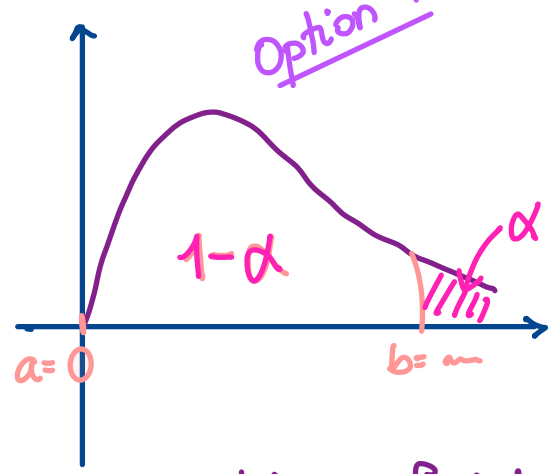
a PIVOTAL QUANTITY for σ^2

Problem 16.5. *Assume that you are assigned a* confidence level $1-\alpha$. *What does it mean to find a* confidence interval *for* $\cancel{\chi^2}$ σ^2?

$$\mathbb{P}[a \leq Q^2 \leq b] = 1-\alpha$$

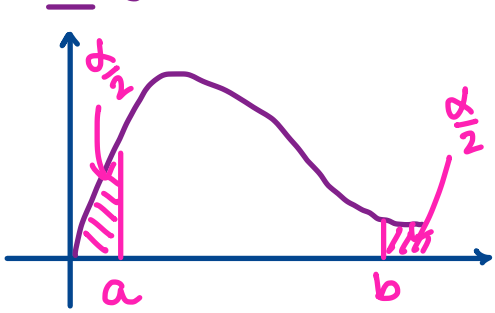
$$\boxed{\mathbb{P}\left[a \leq \frac{(n-1)S^2}{\sigma^2} \leq b\right] = 1-\alpha}$$

Option 1.



Problem 16.6. *Are* $\cancel{\chi^2_L}$ a $\cancel{\chi^2_U}$ b *and as above uniquely defined?*

No :)



We can choose a symmetric conf. int:

a = qchisq(α/2, df = n-1)

b = qchisq(1-α/2, df = n-1)

Problem 16.7. *What's the form of the confidence interval, then?*

$$\Rightarrow \mathbb{P}\left[\frac{1}{a} \geq \frac{\sigma^2}{(n-1)S^2} \geq \frac{1}{b}\right] = 1-\alpha$$

$$\mathbb{P}\left[\underbrace{\frac{(n-1)S^2}{b}}_{\hat{\theta}_L} \leq \sigma^2 \leq \underbrace{\frac{(n-1)S^2}{a}}_{\hat{\theta}_R}\right] = 1-\alpha$$

Problem 16.8. *Assume the above setting. Let the random sample be of size $n = 9$. You do the arithmetic and arrive at the estimate $s^2 = 7.93$ (based on the data set). Using the above procedure, find the 90%−confidence interval for σ^2.*

$\longrightarrow$: $\boxed{n=9}$

$s^2 = 7.93$

$\alpha = 0.10$

Our confidence interval:

$$\left(\frac{8 \cdot 7.93}{\underset{15.51}{b}}, \frac{8 \cdot 7.93}{\underset{2.733}{a}}\right) = \underline{(4.091, 23.216)}$$

$\square$

a = qchisq(0.05, df=8)

b = qchisq(0.95, df=8)